

Development of an EEG-based Motor Imagery Brain-Computer Interface System for Lower Limb Assistive Technologies

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Abstract — Recent studies proved that advancements in Brain-Computer Interface (BCI) have played a vital role for the rehabilitation of people suffering from motor neuron diseases, which affect either the upper limb, lower limb, or both. Despite the fact that significant progress has been done for the field of BCI systems relating to upper-limb impairment, studies supporting BCI for the lower limb are still insufficient. This study aims to develop an Electroencephalogram (EEG)-based Motor Imagery BCI system for lower limb movements usually used for everyday actions (i.e. walking up and down the stairs, and their sub-phases such as stance and swing) which will support the motor-impaired to be able to conduct daily life activities. In line with this, the brain activities were measured via EEG-based BCI system, using a 14-channel EMOTIV EPOC+ device which provided raw signals. The acquired signals were then preprocessed using a built-in denoising filter and the stance and swing gait phases were segmented with the aid of a gyroscope. Each segment underwent feature extraction using Discrete Wavelet Transform (DWT). The interclass correlations of the extracted features were determined. Using the Artificial Neural Network (ANN) as the classification technique with a 10-fold cross validation, an average accuracy of 94.59% was achieved.

Keywords — Motor Neuron Disorders, Brain-Computer Interface, neurerehabilitation, Electroencephalography, Motor Imagery, Emotiv Epoc+, Discrete Wavelet Transform, T-test, Interclass correlation, Artificial Neural Network, k-fold Cross Validation

I. INTRODUCTION

Motor neuron disorders (MNDs) are neurological diseases that severely damage the nerves in the brain and the spine [1] and destroy motor neurons [23].

The brain consists of neurons which act as gateways of inhibitory or excitatory activity. Neurons generate an electrical field which is strong enough to spread through tissue, bone, and skull [4]. Brain-Computer Interface (BCI) could be considered as a pathway for direct communication between the brain and the computer, without the need for any motor movement, by analyzing brain signals [5]. Also, by generating activity-dependent brain plasticity, BCIs aid in restoring motor function [24]. However, BCIs do not necessarily replace a motor function, nonetheless, they may be

used for the patient's rehabilitation to restore or at least improve a particular motor function.

Electroencephalography (EEG) is a non-invasive method utilized to record and document the electrical activities produced by the brain [4]. EEG-based BCIs have attained recognition in clinical practices due to their cost-effectiveness and high temporal resolution in matter of milliseconds.

Motor imagery (MI) is a mental activity in which patients (or study volunteers) are directed to visualize themselves executing a particular motor activity without any visible output of the activity [2].

Previous studies have observed brain electrical activities measured by EEG signals while movements are being imagined [25]. The activation caused by the concerned motor neurons was found to have similarities when the movement is physically executed or just imagined [26]. Different EEG signal patterns are measured depending on the motor imagery executed. The beta and mu rhythms are related with cortical areas which are affixed to the motor output channel of the brain [16]. A decrease of rhythmic activity in the mu and beta bands is usually caused by a motor execution or an intention for movement and labeled as the event-related desynchronization (ERD). The contrary principle of this is the event-related synchronization (ERS) which is an increase in the rhythmic activity and it usually happens during the neutral state after performing a specific motor action. Moreover, the most pertinent for BCI is that ERS and ERD are not only related to the actual execution of a movement. They may also be present during the imagination of movement [20].

BCI research studies have been progressing over the past years relating to upper limb movements. Thus, there are still limited researches regarding BCI development related to lower limb [10], yet significant since recovery for capability to walk is an important rehabilitation aim [9]. A possible reason for the limitation on the studies regarding this is that the brain areas responsible for lower limb movements are located near the *margo superior cerebri* [10]. Therefore, there is a need to perform and associate more beneficial movements (i.e. walking up and down the stairs). This study aims to develop an EEG-based motor imagery BCI system interface for lower

limb assistive technologies, which may aid lower limb movements that are usually performed for everyday actions.

Climbing the stairs is a common daily activity, therefore being able to do it efficiently is important to an individual's daily quality of life. The gait cycle for both stair ascent and stair descent are divided into two phases: the swing phase and the stance phase [25].

Selected male study volunteers performed stair ascent and descent. Using an EMOTIV EPOC+, alongside a gyroscope as a basis for the determination of the stance and swing phases of the right leg, raw EEG signals were acquired from study volunteers. The obtained raw data underwent preprocessing for the removal of unwanted artifacts and segmentation was done to acquire the portions of interest in the raw signal. The segmented signals pertaining to each class was then introduced to the feature extraction stage. In this study, the feature extraction technique that was utilized was the Discrete Wavelet Transform (DWT). For DWT, the Daubechies (Db4) mother wavelet with a decomposition level of 4 was used. The feature vectors per class were compared through the use of the concept of interclass correlation to determine if they are different with one another. To further reduce the number of features, the Principal Component Analysis (PCA) was used. Artificial Neural Network (ANN) was then used as the classifier.

II. METHODOLOGY

A. Experimental Set-Up

For this study, data were gathered from three healthy male volunteers. These volunteers were given an orientation and a pre-study of EEG recording for them to be familiarized with the entire procedure [12]. The volunteers were asked to avoid applying hair products such as hair wax and gel.

Preparation of devices, initialization of software applications, and final reminders for the volunteer were done at least 30 minutes before each session. One session was divided into two parts: the stair ascent and the stair descent. Each part of the session was conducted 10 times. Before and after each trial, volunteers were asked to blink to help indicate the start and end of the execution phase. A 10-second begin-ready signal was given before being asked to walk-up or walk-down a 4-level stair. After each trial, a 30-second break was given wherein volunteers were allowed to move or relax freely and were then reminded to return to their neutral state in preparation for the next trial. Using a stair with 4 levels, a trial for stance and for swing were gathered in each activity. Each class (up-stance, up-swing, down-stance, and down-swing) has 10 trials from the conducted session.

B. Data Acquisition

The acquisition of data was done under a controlled environment for several factors may affect the results of the data acquisition. Noise was kept at minimum, and lighting and temperature were kept at a comfortable range [13][14]. This controlled environment was maintained consistent throughout all the sessions.

The study considered four lower limb motor classifications (the swing and the stance phases of walking down the stairs on the right foot, and the swing and the stance phases of walking up the stairs on the right foot). For the entire procedure, the volunteers wore a 14-channel Emotiv EPOC+ to harvest raw EEG signals. Each electrode of the device was arranged in a cap according to the sensor placement system of 14-channel Emotiv EEG seen in Figure 1. The EEG must be of at least 90% stable contact quality. Among the 14 channels, only four channels were considered, namely: F3 (CH3) which is responsible for motor planning, F4 (CH12) responsible for motor planning for the left upper body, FC5 (CH4) which is the right body controller, and FC6 (CH11) which controls the left part of the body [15].

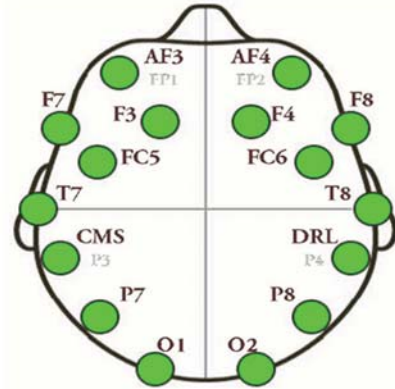


Figure 1. Electrode Placement for the EMOTIV EPOC+ system

A 3-axis Gyroscope connected to a controller, and was programmed using Arduino, was also placed along the shank of the volunteer to measure the shank angular velocity in the sagittal plane (X-axis) [16]. Its data was collected using the Parallax Data Acquisition tool (PLX-DAQ) software add-on for Microsoft Excel.

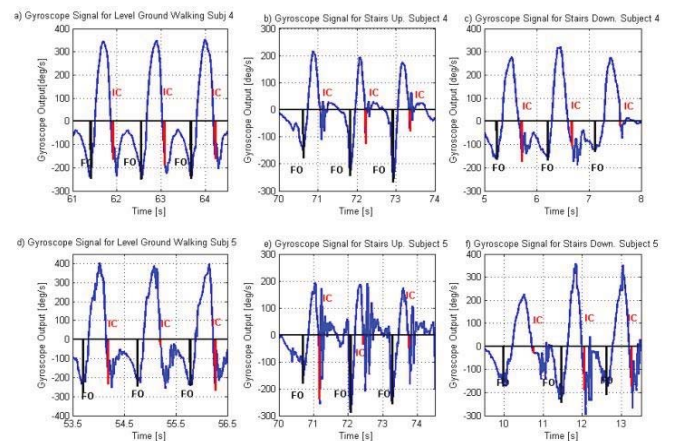


Figure 2. Sample gyroscope signals from [17]

The gyroscope signal recorded from the shank angular velocity was plotted with the recorded EEG signals to determine when the different phases of stair gait occur. As shown in Figure 2, the swing phase of the gait cycle is the

highest positive wave. The determination of initial contact and foot off events (stance phase) is based on of two negative peaks before and after the swing phase. The initial contact is determined by the first negative peak after the swing phase. Meanwhile, the first negative peak after the swing phase is the foot off [17].

C. Preprocessing

The fact that the recorded brain signals have very small amplitude allows minimal movements to affect the EEG signal and to produce a signal with a low signal to noise ratio (SNR). This introduces a possibility of error in decoding the user thoughts [7].

The preprocessing stage, as the first stage, is essential for the performance of the BCI based application. Preprocessing includes noise reduction of the signal, filtering, and segmentation which aids in improving the system's performance by separating the unwanted noise from the actual signal [8]. The EEG signal behaves very randomly — creating a great variability among different subjects and even between the trials of the same subject.

To acquire the desired stimulus-related EEG data from the whole data, segmentation was done. The data from the gyroscope was used as the basis of the segmentation of the generated EEG signal from the Emotiv EPOC+.

D. Feature Extraction Technique

Feature extraction includes the reduction of dimensionality of the dataset and elimination of redundant data [6]. Discrete Wavelet Transform (DWT) recursively subsamples the input signal to reach the desired number of decomposition levels. For each level, the signal is filtered using a low and high-pass filter. The low-passed signal contains the approximation coefficients while the high-passed signal contains the detailed coefficients [19].

For this study, Daubechies 4 (Db4) was used with a 4-level DWT to extract the features. Db4 wavelet provides more details than the other mother wavelets and that it is the most suitable to process biomedical signals [18]. Detail Coefficients cD2 and cD3 were extracted as they have the frequency range which is within the range of the beta band (16-32 Hz) and the mu band (8-16 Hz) respectively, where the features (ERD/ERS) of motor imagery EEG appear. The size of the feature vector was also reduced by determining the statistical parameters of the all wavelet coefficients in each sub-band like:

1. Mean of the absolute values,
2. Standard deviation,
3. Average power,
4. Energy,
5. Kurtosis,
6. Skewness, and
7. The three Hjorth parameters which indicate the statistical property of a signal in time domain:
 - a. Activity (Variance),
 - b. Complexity, and
 - c. Mobility

E. Feature Dimensionality Reduction

Principal Component Analysis (PCA) is a method that can be used to compress and classify data. It reduces the dimensionality of the samples by generating a new set of variables of smaller size than the original set that contain most of the samples' information [19]. The essential information from the new set are called principal components [20].

By applying PCA to the input feature matrix F , the projection matrix W was obtained. Components that contain 98% of variance in the data were utilized. The equation:

$$S = FW$$

was used to obtain the reduced feature matrix S [19].

F. T-Test

T-test is being used in statistics to help weigh the significant difference of two groups using their means [23]. It allows to test an assumption that is applicable to the concerned population.

The t-test was performed by individually calculating the p-values of each of the 4 classes (stance during stair ascent, swing during stair ascent, stance during stair descent, and swing during stair descent) to the neutral state. A p-value below 0.05 is significant, while a p-value above 0.05 is not.

G. Interclass Correlation

Interclass correlation measures the relation of the average feature sets from two different classes. The interclass correlation coefficient is expected to have a value close to 0 to indicate that the two classes are not associated to one another.

H. Artificial Neural Network (ANN)

The classification technique used in this study was the Artificial Neural Networks (ANNs). This classification technique is designed in such a way that it is similar to a biological neural network [11]. The pattern recognition tool (nprtool) in MATLAB was used to implement ANN. The input was constituted by 7 elements, the discriminative features of a class. The output consists of 5 layers pertaining to the 5 classes: neutral, up-stance, up-swing, down-stance, and down-swing. By getting the sum of the number of output and two-thirds of the number of input, the number of hidden layers was computed. Thus, the number of hidden layers used in the ANN was 10. The feature matrix was divided into 15% testing set, 70% training set, and 15% validation set.

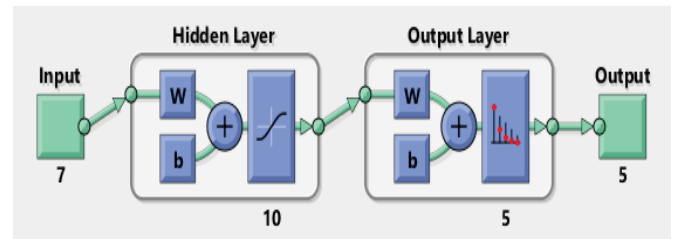


Figure 3. Artificial Neural Network (Feed Forward)

I. K-fold Cross Validation

It is necessary to make the data into subsets in validating the entire input data. K-fold Cross Validation is a statistical technique that divides the given sample data into k subsets. Among those k subsets, $k-1$ subsets will be used for training the proposed system while the remaining subset will be used for testing the performance of the proposed system. The process will be repeated for k number of times until all the subsets were utilized as validation data. Then, the results of the k -folds will be averaged to determine the accuracy of the estimation [21].

In this study, the K-fold Cross Validation will evaluate the performance of the classifier. Specifically, 10-fold cross validation will be utilized to train and test extracted features from the classifier.

In a 10-fold cross-validation, the data set is randomly divided into ten subsets which are then introduced into the classifier. In each fold, using the 10 subsets, 9 subsets will be used to train the classifier while the remaining subset will be used to test the classifier. Each fold will utilize a different subset and will have a resulting percentage. The validation will be done 10 times resulting to 10 percentages. The resulting average percentage is then used to evaluate the performance of the classifier [22].

III. DATA AND RESULTS

The signals from the selected channels were preprocessed and were introduced to the Discrete Wavelet Transform (DWT), which is the feature extraction method used for this study. Daubechies 4 was used with a 4-level DWT to extract the features. Detail Coefficients cD2 and cD3 were extracted which comes in the range of beta and mu bands.

Obtaining the statistical parameters of all of the wavelet coefficients in each sub-band reduces the feature vector dimension. The Mean of the Absolute Values, Standard Deviation, Average Power, Energy, Kurtosis, Skewness, and the three Hjorth parameters which are Activity (Variance), Complexity, and Mobility are the statistical parameters that were calculated. Therefore, feature vectors with 80 elements, 20 for each of the 4 selected EEG channels, were generated.

The dimensions of a composite feature vector are known to be of large amounts and contains redundant features. Therefore, there is a need to further reduce the size of the feature set. After performing PCA, the dimension of the feature vector was reduced from 80 to 7. These features will hence be used for the classification stage.

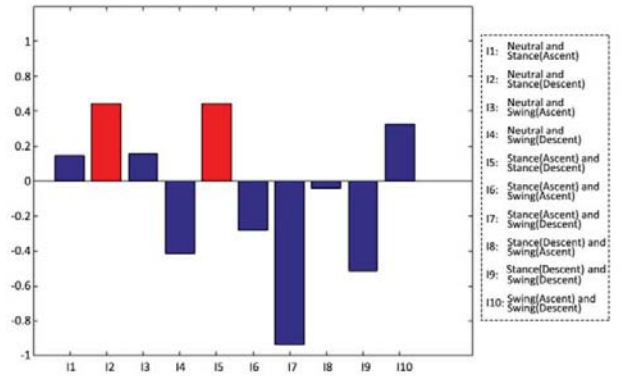


Figure 4. Interclass correlation of two gait phases

From Figure 4, it was observed that most of the interclass correlation coefficients of swing and stance gait phases of all the volunteers are near zero or negative. The ones that are close to zero means that there is no meaningful relationship between the values in the feature sets of the two classes. A negative correlation coefficient, however, means that the values in the feature sets of the two classes tend to move in an opposite direction - when a value of a feature increases in one class, the value of that feature decreases on the other class. In short, they are inversely proportional with one another. It can be observed that the stance and swing phases during stair ascent and stair descent respectively shows a nearly perfect negative correlation.

However, the stance phase during stair descent and stance phase during stair ascent, and the neutral state and stance phase during stair descent have the most positive correlation coefficients which are close to 1. This means that out of all the gait phases, these phases have the highest rate to be misclassified at the classification stage.

In statistics, p-values are widely used. Also known as the calculated probability, p-value is used to evaluate how true a null hypothesis is based from a given or acquired sample data. The pre-chosen probability value, also known as alpha, results to a value of 0.05 in accordance to the majority of studies that uses p-values. Alpha is then used as the cutoff for significance. A p-value of less than 0.05 pertains to an existence of a significant difference, thus, the statement that the null hypothesis has no difference between the means is rejected.

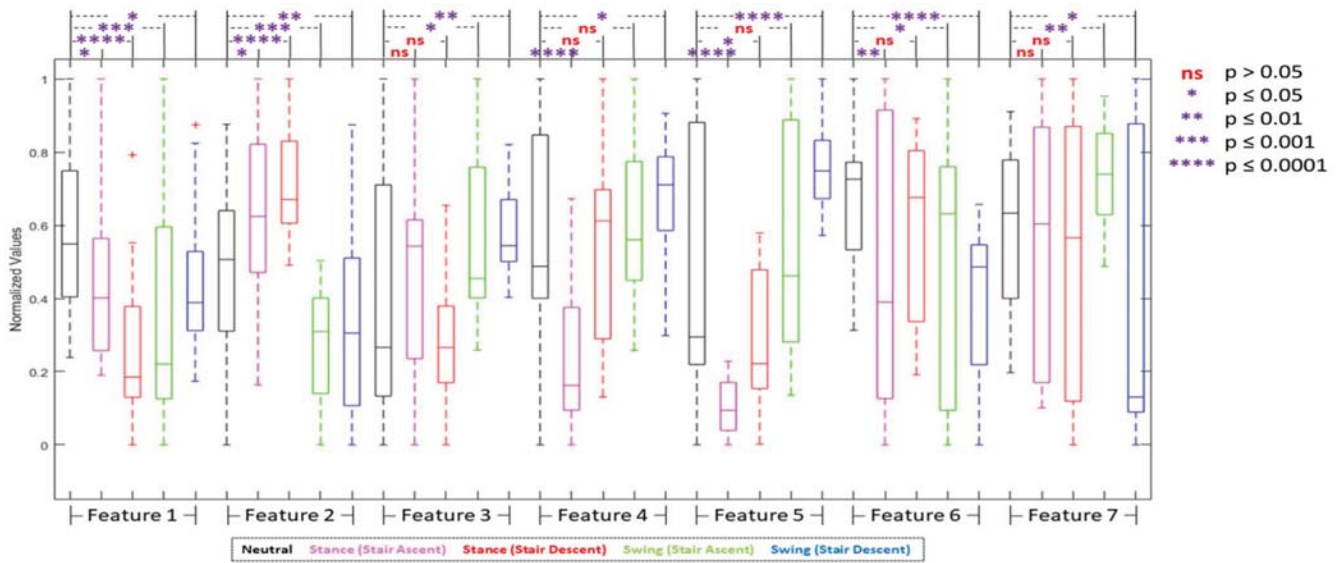


Figure 5. Box Plot of the Reduced Features of the Different Classes for Volunteers 1, 2, and 3

Figure 5 shows the box plots of the reduced features of each class. The p-values are shown with a corresponding legend, based on Figure 5, stating the level of significance. For Volunteers 1, 2, and 3, feature numbers 1 and 2 do not have p-values that are greater than 0.05. This means that these features are the most discriminative and statistically different with one another.

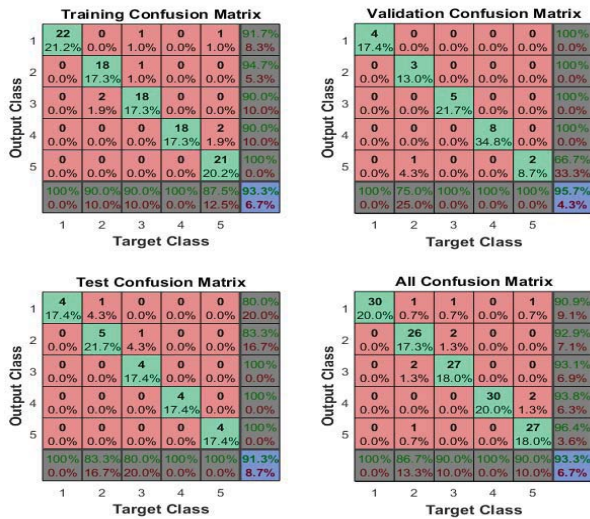


Figure 6. 10th-Fold Cross Validation Confusion Matrix

Figure 6 contains four confusion matrices: training, validation, test and output confusion matrices. Each matrix shows the classification and misclassification of the five classes (1-Neutral, 2-Stance (Ascent), 3-Stance (Descent), 4-Swing (Ascent), and 5-Swing (Descent)).

K-Fold	Train (%)	Validation (%)	Test (%)	Output (%)
1	100	95.7	91.3	98
2	100	95.7	73.9	95.3
3	98.1	95.7	100	98
4	100	91.3	91.3	97.3
5	100	100	91.3	98.7
6	77.9	73.9	69.6	76
7	98.1	95.7	95.7	97.3
8	98.1	91.3	91.3	96
9	98.1	95.7	87.0	96
10	93.3	95.7	91.3	93.3
Average	96.36	93.07	88.27	94.59

Table 1. 10-fold Cross Validation Results

The researchers made use of the 10-fold cross validation. The average output accuracy of 94.59% for the 10 folds was computed by averaging the testing accuracy of each fold. The accuracies for testing, validation, training and output for each fold is tabulated in Table 1. Based on the table, the 6th fold has the lowest testing, validation, training, and output accuracies while the 5th fold has the highest testing, validation, and output accuracies. The accuracy of each fold depends on the training and testing sets. Majority of the folds did not misclassify the phases and validates that the classifier's performance is accurate.

IV. CONCLUSION AND FUTURE WORK

The main objective of this paper is to decode stance and swing phases during stair ascent and descent from the EEG signal, which could later be used to aid for lower limb prosthetic technology designs. Here, the Artificial Neural Network was used to classify these phases. Artificial Neural Network classifier was implemented to obtain distinctive patterns from the EEG signals. It is an algorithm designed to perform like how a human brain processes information. The results were further evaluated using a 10-fold cross validation for the performance of the classifier, and a confusion matrix for the accuracy of the system. The average classification accuracy of 94.59% was achieved.

In the coming days, more volunteers would be included for this study to test the robustness of the system and the classification accuracy would then be recomputed.

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